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# Anemia in Chronic Kidney Disease

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Anemia is an almost universal complication of chronic kidney disease (CKD). It contributes considerably to reduced quality of life (QOL) for patients with CKD and has been associated with a number of adverse clinical outcomes. Before the availability of recombinant human erythropoietin (EPO; epoetin), patients on dialysis frequently required blood transfusions, exposing them to the risks of iron overload, viral hepatitis, and human leukocyte antigen (HLA) sensitization, which reduced the chances of successful transplantation. The advent of erythropoiesis-stimulating agents (ESAs) in the late 1980s changed this situation completely. The ability to correct anemia has not only reduced fatigue and increased physical capacity but also improved a broad spectrum of physiologic functions. Thus, there is a strong rationale for managing anemia in CKD patients, and yet the optimal treatment strategies are still incompletely defined. Apart from therapy with ESAs, iron replacement is essential for anemia management, and thresholds for iron parameters in patients treated with maintenance dialysis must be more liberal than in those with normal kidney function to ensure optimal rates of red blood cell (RBC) production. The costs of anemia management have fallen but are still significant, and because full anemia correction may cause harm, careful consideration of the risks and benefits is mandatory.

## PATHOGENESIS

The anemia associated with CKD is typically an isolated normochromic, normocytic anemia with no leukopenia or thrombocytopenia. Both RBC life span and the rate of RBC production are reduced, but the latter is more important. The normal bone marrow has sufficient capacity to compensate for the reduction in erythrocyte life observed in association with CKD, but this capacity is impaired in CKD. Serum EPO levels remain within the normal range and fail to show the inverse exponential relationship with blood oxygen content characteristic of other types of anemia. EPO is normally produced by interstitial fibroblasts in the kidney cortex, in close proximity to tubular epithelial cells and peritubular capillaries.[1,2] In addition, hepatocytes and perisinusoidal Ito cells in the liver can produce EPO (Fig. 86.1). Hepatic EPO production dominates during fetal and early postnatal life but does not compensate for the loss of kidney production in adult organisms. Subtle changes in blood oxygen content induced by anemia, reduced environmental oxygen concentrations, and high altitude stimulate the secretion of EPO through a widespread system of oxygen-dependent gene expression.[2–4] Central to this process is a family of hypoxia-inducible transcription factors (HIFs), composed of different oxygen-regulated α subunits and a constitutive β subunit. The production of HIF-α is largely independent of oxygen, but its degradation is related to cellular oxygen concentrations. Hydroxylation of specific prolyl and asparagyl residues of HIF-α, for which molecular oxygen is required as a substrate, determines proteasomal destruction of HIF and inhibits its transcriptional activity. Apart from EPO, several hundred HIF target genes have been identified. HIF-2, rather than HIF-1, is the transcription factor primarily responsible for the regulation of EPO production.[5,6] The discovery of the HIF pathway and the underlying mechanisms of oxygen sensing was awarded the Nobel Prize in Physiology or Medicine in 2019.[7]

The role of renal EPO production in the pathogenesis of the anemia associated with CKD is supported by the particularly severe anemia in anephric individuals. However, the mechanisms impairing EPO production in diseased kidneys remain poorly understood. The production capacity for EPO remains significant, even in kidney failure. Thus, patients with anemia and CKD can respond with a significant increase in EPO production to an additional hypoxic stimulus.[1] The main problem therefore appears to be a failure of EPO production to increase in response to chronically reduced hemoglobin (Hb) concentrations. In line with this view, endogenous EPO production can be induced in CKD patients by pharmacologic inhibition of HIF degradation (see later).

EPO is a glycoprotein hormone consisting of a 165–amino acid protein backbone and four complex, heavily sialylated carbohydrate chains.[1] The latter are essential for the biologic activity of EPO in vivo because partially or completely deglycosylated EPO is rapidly cleared from the circulation. This is why recombinant EPO must be manufactured in mammalian cells; bacteria lack the capacity to glycosylate recombinant proteins.

EPO stimulates RBC production by binding to homodimeric EPO receptors, which are primarily located on early erythroid progenitor cells, the burst-forming units erythroid (BFU-e) and the colony-forming units erythroid (CFU-e). Binding of EPO to its receptors salvages these progenitor cells and the subsequent earliest erythroblast generation from apoptosis, thereby permitting cell division and maturation into RBCs.[8] Inhibition of RBC production by yet unknown uremic inhibitors of erythropoiesis may contribute to the pathogenesis of anemia associated with CKD, and dialysis per se can improve the anemia and the efficacy of ESAs. Moreover, the interindividual dose requirements for ESAs vary considerably among CKD patients, and the average weekly dose is much higher than estimated production rates of endogenous EPO in healthy individuals. An alternative view to the accumulation of inhibitors of erythropoiesis in CKD is that in many patients there is overlap between renal anemia and the anemia of chronic disease, which is characterized by inhibition of EPO production and EPO efficacy, as well as by reduced iron availability, mediated through inflammatory cytokines.[9] The hepatic release of hepcidin, the key regulator of iron metabolism, is upregulated in inflammatory states. Hepcidin simultaneously blocks iron absorption from the gut and promotes iron sequestration in macrophages.[10]

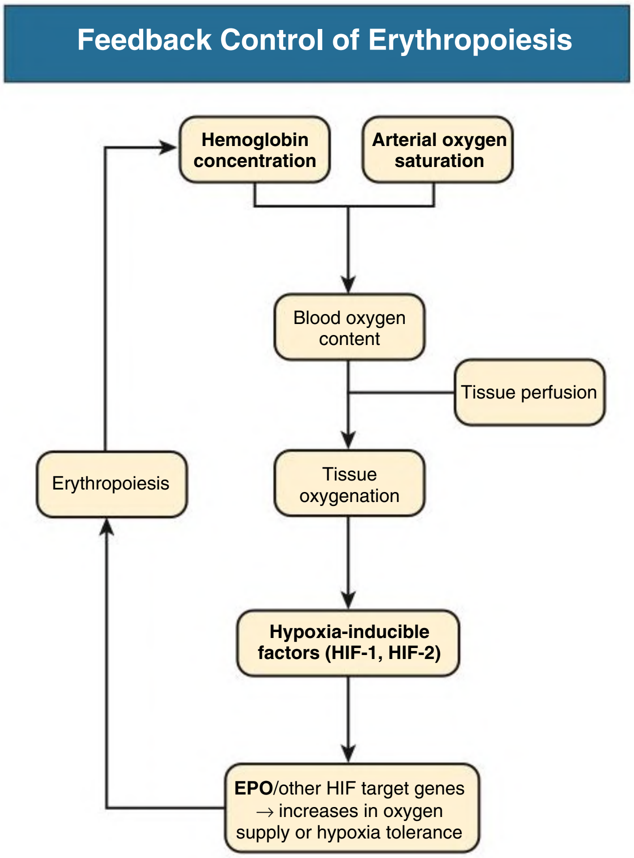


**Fig. 86.1** Feedback control of erythropoiesis. *EPO,* Erythropoietin.

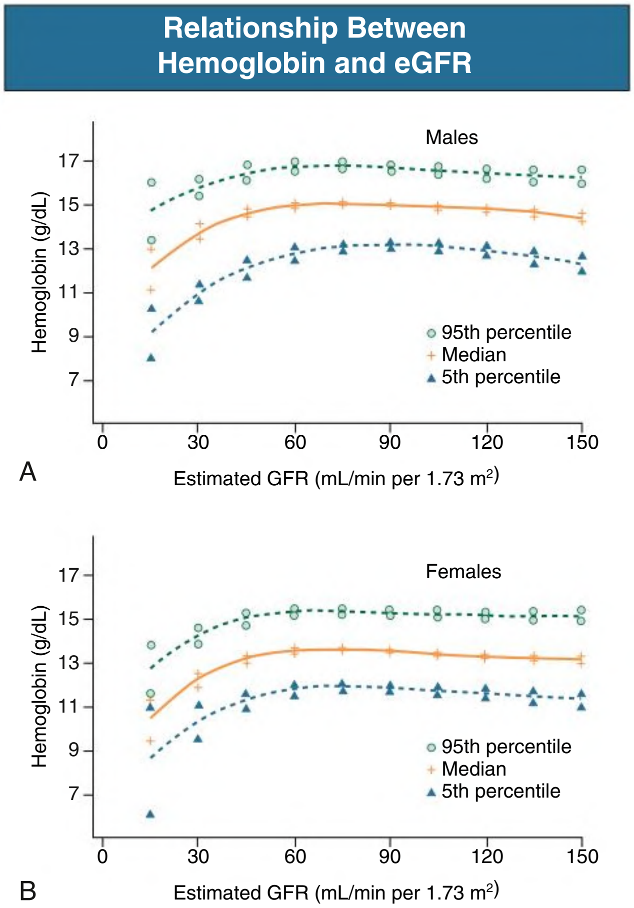


**Fig. 86.2** Relationship Between Hemoglobin Concentration and Estimated Glomerular Filtration Rate (eGFR). Data are from a cross-sectional survey of individuals randomly selected from the general U.S. population (Third National Health and Nutrition Examination Survey [NHANES III]). Results and 95% confidence interval are shown for males (A) and females (B) at each estimated GFR interval. (From McClellan W, Aronoff SL, Bolton WK, et al. The prevalence of anemia in patients with chronic kidney disease. *Curr Med Res Opin.* 2004;20:501–510.)

## EPIDEMIOLOGY AND NATURAL HISTORY

The incidence and severity of anemia progressively increases at lower glomerular filtration rate (GFR). The reported prevalence of anemia by CKD stage varies significantly and depends to a large extent on the definition of anemia and whether study participants are selected from the general population, are at high risk for CKD, have diabetes, or are already under the care of a nephrologist. Data from the National Health and Nutrition Examination Survey (NHANES) showed that the distribution of Hb levels starts to fall at estimated GFR (eGFR) less than 75 mL/min/1.73 m$^2$ in males and less than 45 mL/min/1.73 m$^2$ in females (Fig. 86.2).[11] The prevalence of Hb values less than 13 g/dL increases at eGFR less than 60 mL/min/1.73 m$^2$ in males and less than 45 mL/min/1.73 m$^2$ in females in the general population. Among patients under regular care for CKD, the prevalence of anemia was found to be much greater, with mean Hb values of 12.8 (CKD stages 1 and 2), 12.4 (CKD stage 3), 12.0 (CKD stage 4), and 10.9 g/dL (CKD stage 5).[12] Although anemia develops largely independently of the cause of kidney disease, there are two important exceptions. Patients with diabetes develop anemia more frequently, at earlier stages of CKD, and more severely at a given level of kidney impairment.[13,14] Conversely, in patients with polycystic kidney disease, Hb is on average higher than in other patients with similar severity of CKD, and polycythemia may occasionally develop.

With the advent of ESAs, Hb values in patients with CKD have changed. In particular, in patients on dialysis, average Hb values have steadily increased for many years and then declined again in view of later evidence recommending lower target levels.[13] The average Hb value still varies considerably among countries, reflecting variability in practice patterns (Table 86.1).[14] Moreover, Hb values vary considerably among patients in the same treatment setting and within patients over time, mainly reflecting persistent and time-dependent changes in responsiveness.

## DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

It is best to diagnose anemia and assess its severity by measuring the Hb concentration rather than the hematocrit (Hct). Hb is a stable analyte that is measured directly in a standardized fashion, whereas the Hct is relatively unstable, indirectly derived by automatic analyzers, and lacks standardization. Within-run and between-run coefficients of variation in automated analyzer measurements of Hb are half and one-third those for Hct, respectively.[13]

There is considerable variability in the Hb threshold used to define anemia. According to the Kidney Disease: Improving Global Outcomes (KDIGO) guideline, anemia should be diagnosed at Hb concentrations of less than 13.0 g/dL in males and less than 12.0 g/dL in females,[15] consistent with the World Health Organization (WHO) definition. In children, age-dependent differences in the normal values must be considered. Normal Hb values are increased in high-altitude residents. It is important to note that thresholds for the diagnosis of anemia and evaluation of the causes should not be interpreted as being thresholds for the treatment of the anemia.[15]

TABLE 86.1 **Hemoglobin Levels in Patients on Dialysis**

| | AMONG PATIENTS ON DIALYSIS FOR >180 DAYS | | | AMONG PATIENTS AT START OF DIALYSIS | | |
|---|---|---|---|---|---|---|
| **Country** | ***N*** | **Mean Hb (g/dL)** | **Hb < 11 g/dL (% of patients)** | ***N*** | **Mean Hb (g/dL)** | **Hb < 11 g/dL (% of patients)** |
| Sweden | 456 | 12.0 | 23 | 168 | 10.7 | 55 |
| United States | 1690 | 11.7 | 27 | 458 | 10.4 | 65 |
| Spain | 513 | 11.7 | 31 | 170 | 10.6 | 61 |
| Belgium | 442 | 11.6 | 29 | 213 | 10.3 | 66 |
| Canada | 479 | 11.6 | 29 | 150 | 10.1 | 70 |
| Australia and New Zealand | 423 | 11.5 | 36 | 108 | 10.1 | 70 |
| Germany | 459 | 11.4 | 35 | 142 | 10.5 | 61 |
| Italy | 447 | 11.3 | 38 | 167 | 10.2 | 68 |
| United Kingdom | 436 | 11.2 | 40 | 93 | 10.2 | 67 |
| France | 341 | 11.1 | 45 | 86 | 10.1 | 65 |
| Japan | 1210 | 10.1 | 77 | 131 | 8.3 | 95 |

Mean hemoglobin (Hb) levels and percentage of patients with Hb levels less than 11 g/dL who have been on dialysis therapy for more than 180 days and at the time of starting dialysis, by country. Data are from the Dialysis Outcomes and Practice Patterns Study, Phase II (DOPPS 2) and are derived from 308 randomly selected, representative dialysis facilities. Note that there are marked differences among countries, but at least one-fourth and up to three-fourths of dialysis patients, and, in most countries, more than two-thirds of patients starting chronic dialysis have Hb values less than the recommended lower target of 11 g/dL.

Modified from Pisoni RL, Bragg-Gresham JL, Young EW, et al. Anemia management and outcomes from 12 countries in the Dialysis Outcomes and Practice Patterns Study [DOPPS]. *Am J Kidney Dis*. 2004;44:94–111.

In addition to the Hb value, the evaluation of anemia in CKD patients should include a complete blood count (CBC) with RBC indices (mean corpuscular Hb concentration [MCHC], mean corpuscular volume [MCV]), white blood cell (WBC) count (including differential), and platelet count. Although the anemia associated with CKD is typically normochromic and normocytic, deficiency of vitamin $B_{12}$ or folate may lead to macrocytosis, whereas iron deficiency or inherited disorders of Hb formation (such as thalassemia) may produce microcytosis. Macrocytosis with leukopenia or thrombocytopenia suggests a generalized disorder of hematopoiesis caused by toxins, nutritional deficit, or myelodysplasia. Hypochromia probably reflects iron-deficient erythropoiesis. The absolute reticulocyte count is a useful marker of erythropoietic activity (normal 40,000–50,000 cells/μL of blood).

Iron status tests should be performed to assess the level of iron in tissue stores or the adequacy of iron supply for erythropoiesis. Although serum ferritin is the only available marker of storage iron, several tests reflect the adequacy of iron for erythropoiesis, including transferrin saturation (TSAT), the percentage of hypochromic red blood cells (PHRC), the reticulocyte Hb content (CHr), the MCV, and the MCHC. Storage time of the blood sample may elevate PHRC, and MCV and MCHC are less than the normal range only after long-standing iron deficiency; in clinical practice, TSAT remains the most frequently used parameter.

Anemia in CKD patients may signify nutritional deficits, systemic illness, or other conditions that warrant attention. Moreover, even at modest degrees, anemia reflects an independent risk factor for hospitalizations, cardiovascular (CV) disease (CVD), and mortality.[13] The diagnosis of anemia caused by CKD requires careful judgment of the severity of anemia in relation to the level of GFR and exclusion of other or additional causes. Because there is significant variability in the degree of anemia in relation to the impairment in kidney function, no simple diagnostic criteria can be applied. Causes of anemia other than EPO deficiency should be considered when (1) the severity of anemia is disproportionate to the reduction in GFR, (2) there is evidence of iron deficiency, or (3) there is evidence of leukopenia or thrombocytopenia. Concomitant conditions such as sickle cell disease may exacerbate the anemia, as can drug therapy. For example, inhibitors of the renin-angiotensin system may reduce Hb levels through (1) the direct effects of angiotensin II (Ang II) on erythroid progenitor cells,[16] (2) accumulation of *N*-acetyl-seryl-lysyl-proline (Ac-SDKP), an endogenous inhibitor of erythropoiesis in patients treated with angiotensin-converting enzyme (ACE) inhibitors,[17] and (3) reduction of endogenous EPO production, possibly because of the hemodynamic effects of Ang II inhibition. Myelosuppressive effects of immunosuppressants may further contribute to anemia.[18] The measurement of serum EPO concentrations is usually not helpful in the diagnosis of anemia associated with CKD because there is relative rather than absolute deficiency, with a wide range of EPO concentrations for a given Hb concentration.

## CLINICAL MANIFESTATIONS

In the early clinical trials of recombinant EPO performed in the late 1980s, the mean baseline Hb concentration was about 6 or 7 g/dL, and this progressively increased to about 11 or 12 g/dL under treatment. Patients subjectively felt much better, with reduced fatigue, increased energy levels, and enhanced physical capacity, and there were also objective improvements in cardiorespiratory function.[19] Thus, many of the symptoms previously attributed to the "uremic syndrome" may have been caused by severe anemia associated with CKD (Boxes 86.1 and 86.2). Although the avoidance of blood transfusions and improvement in QOL are obvious early changes, there are also possible effects on the CV system (see Box 86.1). The physiologic consequences of long-standing anemia are an increase in cardiac output and a reduction in peripheral vascular resistance. Anemia is associated with the development of left ventricular (LV) hypertrophy in CKD patients and is thought to exacerbate LV dilation. Sustained correction of severe anemia in CKD patients tends to reverse most of these CV abnormalities, with the notable exception of LV dilation (although anemia

**BOX 86.1 Cardiovascular Effects Resulting From Anemia Correction**

Reduction in high cardiac output
Reduced stroke volume
Reduced heart rate
Increase in peripheral vascular resistance
Reduction in anginal episodes
Reduction in myocardial ischemia
Regression of left ventricular hypertrophy
Stabilization of left ventricular dilation
Increase in whole blood viscosity

**BOX 86.2 Other Effects of Anemia Correction**

**Beneficial**
Reduced blood transfusions
Increased quality of life
Increased exercise capacity
Improved cognitive function
Improved sleep patterns
Improved immune function
Improved muscle function
Improved depression
Improved nutrition
Improved platelet function

**Adverse**
Hypertension
Vascular access thrombosis
Increased risk for stroke

correction may prevent further dilation in some patients[20]). Other effects of anemia correction reported in clinical trials include improvements in QOL, cognitive function, sleep patterns, nutrition, sexual function, menstrual regularity, immune responsiveness, and platelet function. The majority of these trials were not placebo controlled, so the spectrum and extent of possible benefits remain uncertain.

There has been considerable debate about the optimal target range of Hb in CKD patients. A presumed improvement in QOL and expectations of positive effects on CV function and kidney disease progression with increasing Hb concentrations led to a suggested level of greater than 10 to 11 g/dL in all CKD patients.[13,14] However, studies targeting Hb concentrations in the near-normal range found no survival benefit and showed various risks, including increased rates of thromboembolic events, strokes, and possibly death.[21–24] Thus, there is a possible tradeoff among improved QOL, reduced transfusion requirements, and risk for harm (see discussion later), and a target Hb level of greater than 13 g/dL should be avoided.[13,15]

# TREATMENT

## Erythropoiesis-Stimulating Agents and Hypoxia-Inducible Factor Stabilizers

### Epoetin Therapy

Manufacture of recombinant human EPO (epoetin) is achieved by gene transfer into a suitable mammalian cell line such as Chinese hamster ovary (CHO) cells. The early clinical trials were conducted with both epoetin alfa and epoetin beta, both produced in CHO cells. Like the endogenous hormone, both epoetins consist of a 165–amino acid backbone with one *O*-linked and three *N*-linked glycosylation chains. Invariably, however, there are some differences in the glycosylation pattern among different recombinant preparations and the endogenous hormone. “Biosimilar” or “copy” epoetins have been approved in many countries of the world after demonstration of their efficacy and safety in clinical trials.[25]

Before 1998, epoetin alfa in Europe was formulated with human serum albumin, but the latter was replaced with polysorbate 80. Epoetin beta is formulated with polysorbate 20, along with urea, calcium chloride, and five amino acids as excipients. The importance of the formulation of epoetins was highlighted in 2002 when an upsurge in cases of antibody-mediated pure red cell aplasia (PRCA) was associated with the subcutaneous use of epoetin alfa after its change of formulation. Patients affected by this complication develop neutralizing antibodies against both recombinant and endogenous hormone, which result in severe anemia and transfusion dependence.[26,27] The development of neutralizing anti-EPO antibodies was also observed during a clinical trial with an epoetin alfa biosimilar,[28] and a low rate of PRCA also occurs with epoetin beta and darbepoetin alfa. Peginesatide was used to treat patients with EPO-antibody induced PRCA (see below), but since it is no longer available RBC transfusions and immunosuppressive therapy remain the only options.

The epoetins are administered either intravenously or subcutaneously. The earliest clinical trials of the epoetins used intravenous (IV) injections two or three times per week. This was partly because of the short half-life of IV epoetin (6–8 hours)[29] and partly because of the convenience for the patient on dialysis. With use of this regimen, more than 90% of patients show a significant increase in Hb concentration. Good iron management is pivotal for the success of epoetin therapy (see later discussion). Although the bioavailability of subcutaneous epoetin is 20% to 30%, the prolonged half-life after subcutaneous (SQ) administration compared with IV administration allows less frequent injections. Furthermore, the dose required to achieve the same Hb response is about 30% lower with SQ than with IV administration.[29] There appears to be little difference in efficacy among the thigh, arm, or abdomen as injection sites.

### Darbepoetin Alfa

Darbepoetin alfa is a second-generation ESA that is a supersialylated analog of epoetin, possessing two extra *N*-linked glycosylation chains. This property confers a lower clearance rate in vivo, and the elimination half-life of this compound in humans after a single IV administration is 25.3 hours versus 8.5 hours for epoetin alfa. Thus, this agent can generally be given less frequently than the standard epoetins, with administration intervals of once weekly and once every 2 weeks with similar dose requirements.[30] In contrast to the epoetins, dosage requirements for darbepoetin alfa in CKD patients are the same for IV and SQ administration. The conversion factor for switching patients from epoetin alfa or beta to darbepoetin alfa is usually quoted as 200 to 1, but this may vary considerably depending on the population, the dose, and the route of administration.

### Methoxy Polyethylene Glycol–Epoetin Beta

Alternative bioengineering techniques to prolong the half-life of EPO further resulted in the development of methoxy polyethylene glycol–epoetin beta (CERA), a pegylated derivative of epoetin beta with an elimination half-life of around 130 hours when administered either via IV or SQ. Many patients can be treated with once-monthly administration of CERA, which has greater efficacy than once-monthly dosing of darbepoetin alfa when administered via IV to hemodialysis (HD) patients.[31] Once-monthly CERA was also noninferior to shorter-acting ESAs with respect to rates of major adverse CV events or all-cause mortality.[32]

## Adverse Effects of the Erythropoiesis-Stimulating Agents

Adverse effects of ESAs include a moderate increase in blood pressure and an increased rate of thromboembolic events, including vascular access thrombosis. Although these effects probably depend to a large degree on the increase in Hb concentration, ESAs may enhance thrombogenicity and tumor growth in patients with malignancy or exacerbate vascular events in CKD independently of Hb concentration. Thus, the lowest possible doses of ESA should be used to avoid the presumed pleiotropic effects of this class of drugs. Similarly, because no "safe" upper dose level has been determined, it is advisable to avoid repeated dose escalation.[15] In the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT), patients with a history of malignancy had an increased rate of cancer-related deaths when treated with darbepoetin.[24] Although no study has investigated anemia management in patients with CKD and active cancer, ESAs should be used only with great caution in such patients, in particular when cure from cancer is the anticipated outcome.[15] Similarly, in patients with a history of stroke or recent venous thromboembolic event, the benefit-to-risk ratio of ESA therapy should be carefully considered (see later discussion).

## Peginesatide

Peginesatide (previously called Hematide) was an EPO-mimetic peptide that shared the same functional and biologic properties as epoetin and was shown to be an effective treatment for anti-EPO antibody–mediated pure RBC aplasia because of a lack of cross-reactivity with anti-EPO antibodies.[33] Approximately a year after its introduction, peginesatide was withdrawn from the market because of severe hypersensitivity reactions, including fatal anaphylactic events.

## Hypoxia-Inducible Transcription Factor Stabilizers

The HIF stabilizers are competitive inhibitors of HIF prolyl hydroxylases[2,3,34] and asparagyl hydroxylase enzymes involved in the metabolism of HIF and its transcriptional activity. The HIF stabilizers, also referred to as *prolyl-hydroxylase domain protein inhibitors* (PHD inhibitors) cause an increase in endogenous EPO production from both the liver and the diseased, nonfunctioning kidneys.[34,35] Such compounds are orally active, and several of these drugs (e.g., roxadustat, daprodustat, vadadustat, and molidustat) are currently undergoing clinical development.[34,36] The results of phase II trials have shown efficacy in terms of increasing or maintaining the Hb level in both dialysis-dependent and non–dialysis-dependent CKD patients and revealed no evidence of acute adverse effects.[36] Levels of endogenous EPO induced by PHD inhibitors are lower than those observed under IV treatment with ESAs, potentially avoiding dose-dependent toxicity of ESAs. On the other hand, PHD inhibitors are likely to have a range of consequences additional to increasing levels of EPO through activation of other HIF-target genes or interference with other cellular pathways.[2,3,34] Such pleiotropic effects could be beneficial, for example, by improving iron availability or reducing lipid levels.[36] However, given the widespread biologic role of HIF, including its effects on neoangiogenesis and vascular function, the long-term benefit-to-risk relationship is difficult to predict and can be assessed only through rigorous long-term observation. Phase III trials comparing vadadustat with darbepoetin alfa showed noninferiority with respect to CV safety in dialysis-dependent CKD patients[37] but inferiority in non–dialysis-dependent CKD patients.[38] Several trials of a global phase III program comparing roxadustat to epoetin alfa or placebo in dialysis-dependent and non–dialysis-dependent patients have also been published.[39–43] Roxadustat is licensed for use in China, Japan, Chile, and South Korea, but its risk-benefit profile is still debated, and a Food and Drug Administration (FDA) Advisory Committee voted against approval in the United States in July 2021.[44] Daprodustat has shown noninferiority to comparator ESAs in both dialysis and nondialysis patients for both Hb and major adverse CV events (MACE) and a significant improvement in the physical component of a QOL measurement in nondialysis patients compared with placebo.[45,46] However, an FDA Advisory Committee recommended approval of daprodustat only in dialysis patients.[47]

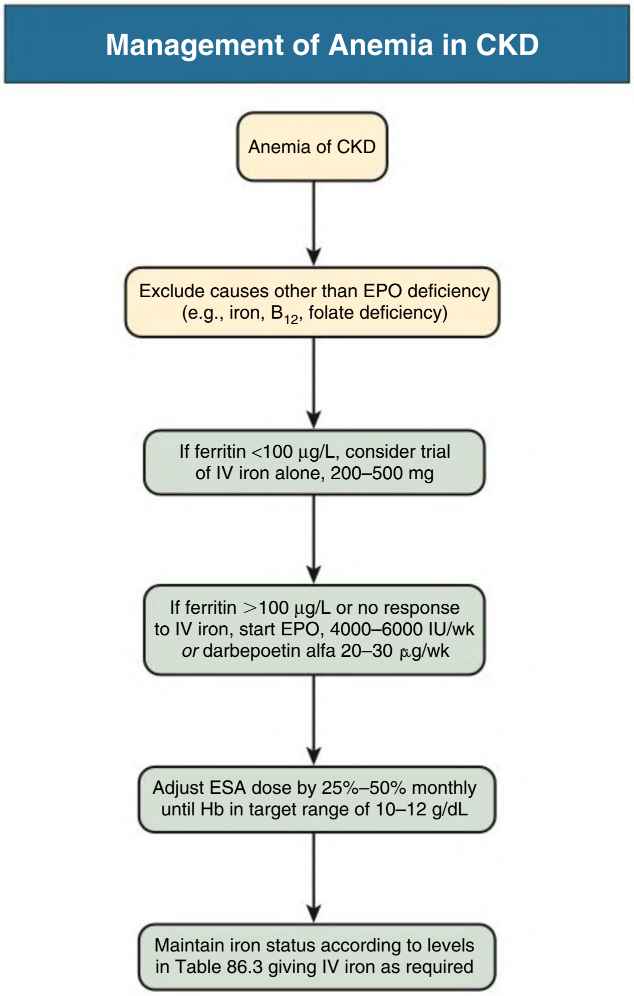


**Fig. 86.3** Management of anemia in patients with chronic kidney disease (CKD). *EPO,* Erythropoietin; *ESA,* erythropoiesis-stimulating agent; *Hb,* hemoglobin; *IV,* intravenous.

## Initiation of and Maintenance Therapy With Erythropoiesis-Stimulating Agents

Before ESA therapy is considered in CKD patients, it is essential to exclude and to correct causes of anemia other than EPO deficiency, such as iron and vitamin deficiencies (Fig. 86.3). If the serum ferritin concentration is less than 100 µg/L, the first therapeutic approach in anemic patients should be iron supplementation. Iron is best given by IV administration, although oral iron can be considered in patients not yet on dialysis.[48] Some patients may respond to IV iron alone (see later discussion). If the ferritin level is greater than 100 µg/L (in the absence of systemic inflammation) or there is a suboptimal response to iron, ESA therapy is an option. However, the Hb level at which ESAs should be initiated remains controversial, mainly because the topic has not been rigorously studied. In TREAT, the largest ESA trial so far, darbepoetin therapy (with a target Hb level of 13 g/dL) was compared with placebo, with a rescue protocol for Hb less than 9 g/dL.[24] In the darbepoetin arm, the number of patients transfused was lower and there was an increase in QOL, but the stroke rate was twice as high, so that the benefit-to-risk relationship was clearly negative. Although these data argue strongly against the initiation of ESA therapy in patients with mild anemia, there is only one, much smaller randomized controlled trial (RCT) in HD patients that tested two different target ranges of 9.5 to 11.0 g/dL and 11.5 to 13.0 g/dL against placebo in patients with severe anemia.[49] Patients in both epoetin treatment arms experienced improved QOL and exercise capacity, but there was no difference between the arms. Based on these findings, the KDIGO guidelines recommend use of ESAs to prevent the Hb level from falling below 9 g/dL.[15] However, individualization is possible, acknowledging the fact that some patients have improved symptoms at higher Hb levels and are prepared to take increased risks. A commonly used target Hb range is between 10 and 12 g/dL. The Hb level should not be intentionally increased to 13 g/dL or more.

The usual IV or SQ starting doses are: for epoetin, about 25 to 50 international units (IU)/kg (e.g., 2000 IU two or three times weekly); for darbepoetin alfa, 20 to 30 μg once weekly; and for CERA, 30 to 60 μg once every 2 weeks. Within 3 to 4 days after treatment initiation, the reticulocyte count typically increases, and within 1 to 2 weeks, there is a significant rise in the Hb concentration, usually of the order of 0.25 to 0.5 g/dL/wk. Thus, over the course of 1 month, a significant increase of 1 to 2 g/dL in Hb concentration is usually achieved. If a patient fails to respond satisfactorily to ESAs, the dose is increased in stepwise upward titrations of 25% to 50%, and if there is still an inadequate response, causes of resistance to ESA therapy should be investigated (see later discussion).

### Hyporesponsiveness to Erythropoiesis-Stimulating Agents

According to KDIGO guidelines, hyporesponsiveness to ESA therapy is identified when the Hb concentration does not increase from baseline after the first month of ESA treatment on appropriate weight-based dosages or if, after treatment with stable doses, patients require two increases in ESA doses up to 50% beyond the dose at which their condition had previously been stable.[15] Patients who are hyporesponsive have a worse prognosis than those who do respond.[50] The causes of resistance to ESA therapy are listed in Box 86.3, and it is important to correct them when possible. The major causes include iron deficiency (see later discussion), infection or inflammation, and underdialysis.[13,15] If the patient is self-administering (e.g., for PD patients), poor compliance with therapy must be excluded. If there is any doubt about the possibility of iron deficiency, a trial of IV iron may be useful. Vitamin $B_{12}$, folate, and thyroxine deficiency can be excluded easily by appropriate laboratory tests, as can severe hyperparathyroidism. Depending on the ethnic origin of the patient, a hemoglobinopathy should be excluded by performing Hb electrophoresis. Some patients taking ACE inhibitors or angiotensin receptor blockers may require higher doses of ESA therapy, although it is rarely necessary to stop these drugs. The possibility of a primary bone marrow disorder, such as myelodysplastic syndrome, should be investigated by a bone marrow examination if all other causes have been excluded. A bone marrow test also may be necessary in the diagnosis of antibody-mediated PRCA, although measurement of the reticulocyte count and anti-EPO antibodies may provide an earlier clue.[26] If a patient receiving ESA therapy has a high reticulocyte count, the bone marrow is generating more than adequate quantities of new RBCs and bleeding or hemolysis should be investigated by means of endoscopy or hemolysis screen (Coombs test, serum bilirubin, lactate dehydrogenase, and haptoglobin levels).

There is no defined upper dose limit of ESAs, and doses of 60,000 IU of EPO per week are not uncommon in the United States, but there is concern that high doses of ESAs may increase side effects independently of Hb concentrations, as described earlier. In patients with acute illness requiring hospitalization, Hb frequently falls despite continued ESA therapy, indicating increased blood loss and temporary hyporesponsiveness. The optimal management of anemia under these conditions remains unclear. Although cost considerations may speak for withholding of ESA therapy until responsiveness is reestablished, it has been proposed that doses should be increased in an attempt to overcome hyporesponsiveness. Anecdotally, very high doses can be effective even in critically ill patients in intensive care units, but an RCT failed to demonstrate a reduction in transfusion requirements and observed an increase in deep vein thrombosis.[51] From a practical point of view, and pending evidence to the contrary, it seems sensible to continue the same dose of ESA during acute illness.

**BOX 86.3 Causes of a Poor Response to Erythropoiesis-Stimulating Agent Therapy**

**Frequent**
Iron deficiency
Infection, Inflammation
Underdialysis

**Less Common**
Poor compliance to ESA therapy
Blood loss
Hyperparathyroidism
Aluminum toxicity (now rare)
Vitamin B12 or folate deficiency
Hemolysis
Primary bone marrow disorders (e.g., myelodysplastic syndrome)
Hemoglobinopathies (e.g., sickle cell disease)
ACE inhibitors, angiotensin receptor blockers
Carnitine deficiency
Anti-EPO antibodies causing pure red cell aplasia

*ACE*, Angiotensin-converting enzyme; *EPO*, erythropoietin; *ESA*, erythropoiesis-stimulating agent.

## Iron Management

Iron is an essential ingredient for heme synthesis, and adequate amounts of this mineral are required for the manufacture of new RBCs. Thus, under enhanced erythropoietic stimulation, greater amounts of iron are used, and many CKD patients (particularly those on HD) have inadequate amounts of available iron to satisfy the increased demands of the bone marrow.[52] Even before the introduction of ESA therapy, many CKD patients were in negative iron balance as a result of poor dietary intake, poor appetite, and increased iron losses from occult and overt blood losses (see Chapter 83). Iron losses in HD patients are up to 5 or 6 mg/day compared with 1 mg in healthy individuals, which may easily exceed the absorption capacity of the gastrointestinal (GI) tract, particularly when there is any underlying inflammation. Iron absorption capacity in patients with CKD is considerably lower than in nonuremic individuals, particularly in the presence of systemic inflammation, and this is mediated by hepcidin upregulation.[10] For this reason also, traditional oral iron therapy (e.g., with ferrous sulfate) is ineffective in many CKD patients, and parenteral iron administration is required, particularly in those receiving HD.[51] However, a newer preparation of oral iron (ferric citrate) that shows greater absorption of iron from the gut is now available in the United States and Japan, and this agent can be used to concomitantly lower phosphate and increase iron levels in CKD patients.

**TABLE 86.2 Recommended Iron Status Levels in Chronic Kidney Disease (CKD)**

| Test | Recommended Range |
|---|---|
| Serum ferritin | CKD: 100–500 μg/L<br>Hemodialysis: 200–500 μg/L |
| Transferrin saturation | 20%–40% |
| Hypochromic red cells | <10% |
| Reticulocyte hemoglobin content | >29 pg/cell |
| Serum transferrin receptor | Not established |
| Erythrocyte zinc protoporphyrin | Not established |

An inadequate supply of iron to the bone marrow may be caused by an absolute or a functional iron deficiency.[51] Absolute iron deficiency occurs when there are low whole-body iron stores, as indicated by a serum ferritin level less than 30 μg/L. Functional iron deficiency occurs when there is ample or even increased storage iron, but the iron stores fail to release iron rapidly enough to satisfy the demands of the bone marrow. Several markers of iron status are available, but none of them is ideal (Table 86.2). Serum ferritin is a marker of storage iron but is spuriously raised in inflammatory conditions and liver disease. TSAT is a function of the circulating serum iron in relation to the total iron-binding capacity and is often regarded as a better measure of available iron; however, levels can be highly fluctuant because of significant diurnal variation in the measurement of serum iron.[52] The percentage of hypochromic RBCs and the CHr are RBC and reticulocyte parameters, respectively, that are indirect measures of how much iron is being incorporated into the newly developing or mature RBCs. No one measure of iron status is usually adequate to exclude iron deficiency, and the recommended levels for these measures are based on limited scientific evidence. Functional iron deficiency is usually diagnosed when there is a normal or increased ferritin level and a reduced TSAT (<20%) or increased hypochromic RBCs (>10%). The KDIGO guideline suggests a trial of iron in CKD patients with anemia who are not on iron and ESA therapy and in whom an increase in Hb concentration is desired and in patients on ESA therapy in whom an increase in Hb concentration or a decrease in ESA dose is intended when TSAT is up to 30% and ferritin level is up to 500 μg/L[15] (see Table 86.2). Levels of ferritin above this threshold usually do not confer any clinical advantage and may exacerbate iron toxicity. The optimal TSAT is greater than 20% to 30% to ensure a readily available supply of iron to the bone marrow. No upper limits of ferritin or TSAT were specified in the KDIGO anemia guideline, largely because there is no robust evidence to determine a threshold beyond which harm or loss of efficacy supervenes. However, until more information become available, the treating nephrologist should exercise caution in administering IV iron to patients with ferritin levels greater than 800 μg/L or TSAT greater than 30%. Several studies support maintaining the percentage of hypochromic RBCs at less than 6% and the CHr at greater than 29 pg/cell.

Oral iron is generally poorly absorbed in people with severe CKD, and there is a high incidence of GI side effects. Thus, IV administration of iron has become the standard of care for many CKD patients, particularly those receiving HD.[52] Several IV iron preparations are available worldwide, including iron dextran, iron sucrose, iron gluconate, and the newer iron preparations ferric carboxymaltose, ferumoxytol, and iron isomaltoside 1000. The last three iron preparations allow higher doses of IV iron to be administered more rapidly, without the need for a test dose. All of the iron preparations contain elemental iron surrounded by a carbohydrate shell, which allows them to be injected intravenously.

An RCT comparing proactive high-dose administration of IV iron sucrose (400 mg iron per month up to a ferritin of 700 μg/L) with reactive low-dose iron sucrose (0–400 mg iron per month) in 2141 incident hemodialysis patients (PIVOTAL)[53] showed lower ESA dose and transfusion requirements, as well as improved mortality and CV events with the high-dose regimen. Follow-up in the study was for a median of 2.1 years (maximum 4.4 years), and thus the longer-term safety of IV iron remains unknown.

All IV iron preparations also carry a risk for immediate hypersensitivity reactions, which may be characterized by hypotension, dizziness, and nausea. These reactions are usually short-lived and caused by too large a dose given in too short a time. Iron dextran also carries the risk for acute anaphylactic reactions because of preformed dextran antibodies, although this was largely a problem with the high-molecular-weight IV iron dextran preparations, which have now been withdrawn from the market.

In nondialysis patients, both oral and IV iron may be used and there are advantages and disadvantages for each route of administration, which have been compared in two RCTs. The FIND-CKD study showed that both routes of administration were effective in this patient population, although higher-dose IV iron resulted in a faster and greater rise in Hb, with no safety concerns.[48] However, the REVOKE study suggested that the rate of CV- and infection-related serious adverse events was greater in the IV iron treatment group compared with the oral iron group.[54] The reasons for these discrepant findings remain unknown. Thus, the physician has to balance the low costs and convenience of oral iron treatment with known poor adherence to treatment, GI side effects, and reduced efficacy; IV iron, on the other hand, requires special facilities and set-up for administration and carries a very small risk for hypersensitivity reactions and potential additional risks.

## SELF-ASSESSMENT QUESTIONS

1. Which of the following statements is *false* regarding the characteristics of renal anemia?
   - **A.** The RBCs produced are usually normochromic and normocytic.
   - **B.** There is usually no associated leukopenia or thrombocytopenia.
   - **C.** The reticulocyte count is usually around 40,000 to 50,000/μL of blood.
   - **D.** Serum erythropoietin levels are usually within the normal range.
   - **E.** The RBC life span is usually normal.
2. Which of the following statements is *false*?
   - **A.** Dose requirements of epoetin are generally 20% to 30% less when the agent is administered subcutaneously compared with intravenously.
   - **B.** ESA therapy should be used with caution in patients with previous or current malignancy.
   - **C.** Patients who are hyporesponsive to ESA therapy have a worse prognosis than those who do respond.
   - **D.** The defined upper dose limit of epoetin is 60,000 IU/wk because it is known that CV toxicity occurs above this dose level.
   - **E.** ACE inhibitors may confer resistance to ESA therapy in some patients.
3. Which of the following statements is *false* regarding the TREAT study?
   - **A.** Patients receiving darbepoetin alfa were randomized to either a target hemoglobin (Hb) of 13 or a target Hb of 9 g/dL.
   - **B.** Patients randomized to a target Hb of 13 g/dL had a small increase in quality of life compared with the control arm.
   - **C.** There was a significant reduction in the use of RBC transfusions in patients randomized to a target Hb of 13 g/dL.
   - **D.** There was a doubling of the rate of stroke in patients randomized to a target Hb of 13 g/dL.
   - **E.** There was a significant increase in cancer-related mortality in the subset of patients with a history of malignancy who were randomized to a target Hb of 13 g/dL.
4. Which of the following statements is *true* regarding IV iron supplementation?
   - **A.** All IV iron preparations require a test dose before their first administration, as per their product label.
   - **B.** All the newer IV iron preparations (ferumoxytol, ferric carboxymaltose, and iron isomaltoside 1000) have the advantage that up to 1 g may be administered as a single dose.
   - **C.** IV iron preparations vary in their lability and speed of iron release from the carbohydrate shell, with iron gluconate being the most stable and iron dextran the least stable compound.
   - **D.** IV iron may improve the anemia of chronic kidney disease in up to 30% of patients not receiving ESA therapy who have a low ferritin level.
   - **E.** Hypersensitivity reactions are more common with low-molecular-weight iron dextrans than with high-molecular-weight iron dextran compounds.

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